Newman-Penrose Formalism Calculations

Stephani et al., Equation (14.23)

Metric

$$ds^2 = -c^2 \sinh(\xi)^{10}\, \mathrm{d}\xi \otimes \mathrm{d}\xi + \frac{1}{9}\, c^2 \sinh(\xi)^{10}\, \mathrm{d}y \otimes \mathrm{d}y + \frac{8}{81}\, c^2 \sinh(\xi)^8\, \mathrm{d}x \otimes \mathrm{d}x + \frac{8}{81}\, c^2 \cosh(y) \sinh(\xi)^8\, \mathrm{d}x \otimes \mathrm{d}z$$
$$+ \frac{8}{81}\, c^2 \cosh(y) \sinh(\xi)^8\, \mathrm{d}z \otimes \mathrm{d}x + \left(\frac{1}{9}\, c^2 \sinh(\xi)^{10} \sinh(y)^2 + \frac{8}{81}\, c^2 \cosh(y)^2 \sinh(\xi)^8\right) \mathrm{d}z \otimes \mathrm{d}z$$

Coordinates

$$(x^\mu) = (\xi, y, x, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\, \sqrt{2} c \sinh(\xi)^5\right) d\xi + \left(\frac{1}{6}\, \sqrt{2} c \sinh(\xi)^5\right) dy$$

$$n = \left(\frac{1}{2}\, \sqrt{2} c \sinh(\xi)^5\right) d\xi + \left(-\frac{1}{6}\, \sqrt{2} c \sinh(\xi)^5\right) dy$$

$$m = \left(\frac{2}{9} i\, c \sinh(\xi)^4\right) dx + \left(\frac{1}{6}\, \sqrt{2} c \sinh(\xi)^5 \sinh(y) + \frac{2}{9} i\, c \cosh(y) \sinh(\xi)^4\right) dz$$

$$\overline{m} = \left(-\frac{2}{9} i\, c \sinh(\xi)^4\right) dx + \left(\frac{1}{6}\, \sqrt{2} c \sinh(\xi)^5 \sinh(y) - \frac{2}{9} i\, c \cosh(y) \sinh(\xi)^4\right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}}\right)\,d\xi + \left(\frac{3\,\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}}\right)\,dy$$

$$n = \left(-\frac{\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}}\right)\,d\xi + \left(-\frac{3\,\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}}\right)\,dy$$

$$m = \left(\frac{9i}{4\,c\sinh\left(\xi\right)^{4}} - \frac{3\,\sqrt{2}\cosh\left(y\right)}{2\,c\sinh\left(\xi\right)^{5}\sinh\left(y\right)}\right)\,dx + \left(\frac{3\,\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}\sinh\left(y\right)}\right)\,dz$$

$$\overline{m} = \left(-\frac{9i}{4\,c\sinh\left(\xi\right)^{4}} - \frac{3\,\sqrt{2}\cosh\left(y\right)}{2\,c\sinh\left(\xi\right)^{5}\sinh\left(y\right)}\right)\,dx + \left(\frac{3\,\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}\sinh\left(y\right)}\right)\,dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}}\,\partial_{\xi} + \frac{3\,\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}}\,\partial_{y}$$

$$\Delta = -\frac{\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}}\,\partial_{\xi} - \frac{3\,\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}}\,\partial_{y}$$

$$\delta = \frac{9i}{4\,c\sinh\left(\xi\right)^{4}} - \frac{3\,\sqrt{2}\cosh\left(y\right)}{2\,c\sinh\left(\xi\right)^{5}\sinh\left(y\right)}\,\partial_{x} + \frac{3\,\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}\sinh\left(y\right)}\,\partial_{z}$$

$$\overline{\delta} = -\frac{9i}{4\,c\sinh\left(\xi\right)^{4}} - \frac{3\,\sqrt{2}\cosh\left(y\right)}{2\,c\sinh\left(\xi\right)^{5}\sinh\left(y\right)}\,\partial_{x} + \frac{3\,\sqrt{2}}{2\,c\sinh\left(\xi\right)^{5}\sinh\left(y\right)}\,\partial_{z}$$

Spin Coefficients

$$\rho = \frac{9\sqrt{2}\cosh(\xi)}{4\,c\sinh(\xi)^6} - \frac{3\sqrt{2}\cosh(y)}{4\,c\sinh(\xi)^5\sinh(y)}$$

$$\sigma = \frac{\sqrt{2}\cosh(\xi)}{4\,c\sinh(\xi)^6} - \frac{3\sqrt{2}\cosh(y)}{4\,c\sinh(\xi)^5\sinh(y)} - \frac{i}{c\sinh(\xi)^6}$$

$$\kappa = 0$$

$$\mu = -\frac{9\sqrt{2}\cosh(\xi)}{4\,c\sinh(\xi)^6} - \frac{3\sqrt{2}\cosh(y)}{4\,c\sinh(\xi)^5\sinh(y)}$$

$$\lambda = -\frac{\sqrt{2}\cosh(\xi)}{4\,c\sinh(\xi)^6} - \frac{3\sqrt{2}\cosh(y)}{4\,c\sinh(\xi)^5\sinh(y)} + \frac{i}{c\sinh(\xi)^6}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{5\sqrt{2}\cosh(\xi)}{4\,c\sinh(\xi)^6} + \frac{i}{2\,c\sinh(\xi)^6}$$

$$\gamma = \frac{5\sqrt{2}\cosh(\xi)}{4\,c\sinh(\xi)^6} - \frac{i}{2\,c\sinh(\xi)^6}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{10\ \cosh\left(\xi\right)^{2}}{c^{2}\sinh\left(\xi\right)^{12}} + \frac{7}{2\,c^{2}\sinh\left(\xi\right)^{12}}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{5\ \cosh\left(\xi\right)^{2}}{c^{2}\sinh\left(\xi\right)^{12}} + \frac{7}{4\,c^{2}\sinh\left(\xi\right)^{12}}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{10\ \cosh\left(\xi\right)^{2}}{c^{2}\sinh\left(\xi\right)^{12}} + \frac{7}{2\,c^{2}\sinh\left(\xi\right)^{12}}$$

$$\Lambda = \frac{13}{3\,c^{2}\sinh\left(\xi\right)^{10}} + \frac{15}{4\,c^{2}\sinh\left(\xi\right)^{12}}$$

Weyl Tensor Components

$$\Psi_0 = -\frac{2}{c^2 \sinh(\xi)^{10}} - \frac{3i\sqrt{2}\cosh(\xi)}{2\,c^2 \sinh(\xi)^{12}} - \frac{3}{2\,c^2 \sinh(\xi)^{12}}$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{2\,\cosh(\xi)^2}{3\,c^2 \sinh(\xi)^{12}} - \frac{i\sqrt{2}\cosh(\xi)}{2\,c^2 \sinh(\xi)^{12}} + \frac{1}{6\,c^2 \sinh(\xi)^{12}}$$

$$\Psi_3 = 0$$

$$\Psi_4 = -\frac{2}{c^2 \sinh(\xi)^{10}} - \frac{3i\sqrt{2}\cosh(\xi)}{2\,c^2 \sinh(\xi)^{12}} - \frac{3}{2\,c^2 \sinh(\xi)^{12}}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type ”D”